Previously Published / Preprint Statement

Hemanth Manchabale Papachappa Aniriuddha Ganguly

September 2026

Manuscript title: Geodesic Diagnostic Trajectories (GDT): A Differential Geometry Framework for Adaptive Decision Support and Operational Optimization in EHR Systems

The authors confirm that this manuscript has **not** been published in whole or in part in any peer-reviewed journal and is **not** currently under review at any other journal.

Related Prior Work by the Authors

The following prior publications and preprints are related to but substantively distinct from the present submission:

1. **Manchabale Papachappa, H., Kumar, A., and Badam, N.** "Riemannian Intent Manifolds for Session-Based Search: A Joint Embedding Predictive Architecture for Intent Forecasting." 2026 (preprint).
   Relationship: This preprint introduces the general JEPA-based intent forecasting framework on Riemannian manifolds for web-scale session search. The present GDT manuscript extends that work to the clinical EHR domain, adds the geodesic shift interference gate, proves Proposition 1 (context interference bound), and reports a four-arm clinical benchmark on MIMIC-III/IV, eICU, and HiRID. The clinical benchmark, formal interference bound, confidence-scaling mechanism, and queueing model are new contributions not present in the preprint.

2. **Manchabale Papachappa, H.** "Continuum Memory Architecture (CMA) for Clinical Search: A Simulation-Based Evaluation of Intent-Aware Retrieval Using Complex Public Clinical Data." SSRN preprint, DOI: 10.2139/ssrn.7197862, 2026.
   Relationship: CMA is a direct predecessor of GDT and serves as the primary comparison system in the four-arm benchmark reported here. GDT differs from CMA in three concrete ways: (1) GDT adds a confidence-scaling mechanism to the JEPA prefetch engine (CMA uses fixed-weight prefetch); (2) GDT formalizes and proves the context interference bound as Proposition 1 (CMA does not); (3) GDT introduces the M/M/c queueing model for operational throughput projection (not present in CMA). These differences are fully described and quantified in the manuscript.

Preprint Policy Compliance

Per JTEHM policy, authors are permitted to post preprints prior to submission. Should the authors post a preprint version of this manuscript to arXiv or a comparable server during the review period, the journal will be notified immediately. The preprint will clearly identify itself as a manuscript under review and will not be promoted in a manner that could compromise the integrity of the peer-review process.

Author signatures:

Hemanth Manchabale Papachappa ______________________________

Aniriuddha Ganguly ______________________________

Date: September 2026